

Joseph Dwyer

916-755-9100 | jdwyer@csus.edu | Joeyyoungdwyer@gmail.com | Antelope, CA | linkedin.com/in/joseph-dwyer-10a4131ba
Mechanical Engineering Candidate | Automation, Robotics, Motion Control & Manufacturing Equipment Support

Professional Profile

Mechanical Engineering candidate focused on industrial automation, robotics, motion control, and manufacturing equipment support. Hands-on experience with ABB/Rexroth robotics coursework, self-hosted AI/Industry 4.0 automation infrastructure, Klipper-based motion systems, CAD/CAM, CNC/laser/3D-printer equipment maintenance, lab test setup, data collection, technical documentation, troubleshooting, and continuous improvement. Strong fit for automation support, commissioning assistance, equipment fault isolation, prototype development, and cross-functional manufacturing environments.

CORE SKILLS

Controls & Automation: ABB/Rexroth robotics coursework, motion control, sensors, low-voltage troubleshooting, commissioning support, fault isolation, root-cause analysis, equipment uptime

Digital Manufacturing / OT: PLC/HMI/SCADA fundamentals, industrial controls terminology, equipment integration concepts, OT/networking basics, Tailscale, Linux, local AI workflow automation, human-reviewed agents

CAD/CAM & Fabrication: Fusion 360, SolidWorks, AutoCAD, CNC routing, Haas CNC, milling, lathe, laser cutting, deburr, rapid prototyping, 3D printing, technical plastics and printer tuning

Programming & Data: Python, C, Java, MATLAB, Klipper, llama.cpp, Qwen/Gemma local models, test data collection, engineering documentation, spreadsheet/test logs

Communication & Support: technical training, operator/user support, vendor-style coordination, shop safety, procedure writing, project planning, sponsorship/outreach, team leadership

EDUCATION

California State University, Sacramento | B.S. Mechanical Engineering | Dec. 2024 - Present | **Expected Fall 2026**

- GPA: 3.358; relevant focus areas include robotics, controls-adjacent automation, manufacturing systems, CAD/CAM, test methods, and mechanical design.

Sierra College | A.S. Natural Science and A.S.-T Engineering | Jul. 2019 - Dec. 2024

- GPA: 2.73; Worked 30-40 hrs/week while completing degree requirements; earned SOLIDWORKS Foundations: Sketching and Extrusion certification.

AUTOMATION, CONTROLS & ENGINEERING PROJECTS

Local AI / Industry 4.0 Automation Server | Independent Project | 2025-Present

- Designed and assembled a self-hosted AI automation server using enterprise hardware, Linux, llama.cpp, Qwen/Gemma-family models, Tailscale networking, and a custom 3D-printed chassis.
- Developing Hermes/Odysseus agent workflows for engineering-team support: email triage, human-reviewed draft generation, translation, meeting-minute generation, calendar support, technical research summaries, and task review.
- Building local-first automation infrastructure to keep engineering context private while supporting deep research reporting, documentation assistance, scheduling support, and prototype decision-making.
- Exploring supervised digital-manufacturing workflows including slicer preparation, 3D-printer job scheduling, print-status monitoring, future vision-based failure detection, and robotic part-handling concepts.

ABB / Rexroth Robotics Coursework | Sacramento State ME165 | 2025-2026

- Gained hands-on academic experience with ABB robot systems and Rexroth controller platforms, including robot motion concepts, controller interaction, lab setup, troubleshooting, and automation-system documentation.
- Applied robotics, controls, and manufacturing-system concepts to equipment-support scenarios relevant to commissioning, operator support, fault isolation, technical manuals, and continuous improvement.
- Connects directly to industrial robot workcell concepts, controller interfaces, electromechanical troubleshooting, and safe lab execution in a manufacturing-style environment.

Custom Motion-Control & 3D Printing Systems | Independent Engineering Projects | May 2019-Present

- Built and troubleshot Voron 2.4, Prusa Core One, and custom CoreXY printer systems using Klipper, stepper motion control, firmware configuration, CAN/networked electronics, hotend/toolhead integration, and custom profiles.
- Designed and modified electromechanical assemblies for rapid prototyping while diagnosing mechanical, thermal, electrical, software, material-flow, and calibration issues across full machine systems.
- Created printer profiles and machine configurations to support high-speed printing, larger-format prototyping, and technical material development for engineering-grade parts.

PROFESSIONAL EXPERIENCE

Lab Assistant | University Enterprises Inc. / Sacramento State | May 2026-Present | Sacramento County, CA

- Support CEC-related appliance energy-efficiency audits, portable spa testing, and residential fan airflow/CFM testing aligned with HVI 916 procedures.
- Build controlled test environments, set up test equipment, collect flow-rate data, maintain test repeatability, and document procedures/results for engineering evaluation.
- Apply hands-on measurement, instrumentation, test setup, and technical documentation habits relevant to manufacturing equipment validation and process improvement.

Chief Design Officer | Metuned LLC | July 2023-Jan. 2026 | Sacramento, CA

- Co-founded racecar electronics startup focused on affordable data-acquisition/performance solutions; supported prototype planning, testing, mechanical/electrical design direction, and product documentation.
- Contributed to concept development, prototype iteration, low-cost manufacturability decisions, and technical planning for adaptable racecar electronics solutions.

Makerspace Technical Assistant | Sierra College | Apr. 2023-Dec. 2023 | Rocklin, CA

- Trained users on safe operation of 3D printers, CNC routers, laser cutters, woodworking tools, embroidery/vinyl equipment, and shop equipment while enforcing safety standards.
- Maintained, organized, and troubleshot makerspace equipment; supported workshops and student projects while improving CAD, manufacturing, teaching, and equipment-support skills.

Inventory Manager / Packaging / Deburr Technician | Vanquish Products / ASB Products | July 2021-Sept. 2023 | Sacramento, CA

- Managed inventory flow for production parts, coordinated anodizing/laser-etch processing, and supported shipping, packaging, deburr, quality, and spare/production part organization.
- Operated and supported shop processes for cleaning, polishing, deburring, production quality, and flexible manufacturing support during high-demand periods.

LEADERSHIP

President / Vice President / Treasurer | Sierra College Motorsports Club | May 2023-Dec. 2024

- Led race-preparation planning for a 24 Hours of Lemons Mini Cooper build; coordinated student tasks, managed sponsorship/outreach, and improved team funding through events and industry partnerships.
- Built practical project-management experience through scheduling, task assignment, team communication, real-time race support, funding strategy, and technical decision-making under pressure.
- Assisted in building the mini cooper to race specification standards and assisted in repairs and troubleshooting the vehicle to maintain track time for drivers and long term reliability for the school to use for the foreseeable future

ADDITIONAL RELEVANT TECHNICAL EXPOSURE

- Comfortable reading technical manuals, documenting procedures, training users, and translating engineering requirements into shop-floor or lab execution steps.
- Hands-on construction internship exposure to framing, plumbing, electrical, landscaping, jobsite safety, tool use, and cross-trade coordination.
- Known strengths: fast learning on unfamiliar equipment, practical troubleshooting across mechanical/electrical/software boundaries, and strong ownership of prototype-to-test cycles.